# Learning from Earlier Websites: Cross-Task Long-Term Memory for a Small Website-Generation Model




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This thesis is for examination purposes only and is confidential to the examination process.

# Abstract

Agents that generate websites with large language models solve every task from scratch: what went wrong on one website is forgotten before the next one starts. This dissertation asks whether a system built around a *small* generator, WebGen-LM-7B, can improve across a sequence of WebGen-Bench tasks when it is given a long-term memory built from its own earlier attempts, without changing its weights or the official evaluation protocol. The memory is maintained by a separate, larger model; only the generator is small.

We first reproduce the official No Memory baseline on a self-hosted cluster (28.5% accuracy on 101 tasks versus 28.4% reported). Its failures show that many zero-score tasks are blank pages caused by basic code errors, such as importing a package that is not installed, and that the model rarely follows advice written as text. A text-based memory therefore helps early in the sequence but falls behind No Memory later, as generic advice accumulates.

Code memory adds a requirement checklist and a code pool: when earlier tasks show a code error that breaks a site, the prompt carries a short fragment copied verbatim from the chain's own compiled, successful websites. In a single run over all 101 tasks it reaches 40.0% against 28.5% (paired gain 11.5 points, 95% CI 6.0–16.9) and cuts zero-score tasks from 30 to 13; an ablation attributes most of the gain to memory rather than to the checklist. Under shuffled task orders, a whole-file example made the generator return the untouched starter project. The final version (v9.1) shows only verified lines of such files; it removes these failures in both shuffled orders and keeps gains of 7.4 and 5.5 points. [TODO: Add v9.1 on the manifest order.] Gains are in functional accuracy, largely from sites that now render; appearance on sites that work in both arms is similar.

The main observation is that, for this small generator, the *form* of memory mattered: short code fragments were followed where textual lessons were not. All results are single runs on one benchmark on which the method was also developed; we discuss these limits, the cost of the larger memory model, and the ethical considerations of code agents that change their behaviour from experience. [TODO: Status: fixed-prefix ablation queued; its result decides whether the gain is described as accumulated memory or as automatically discovered prompt code.]

# Acknowledgements

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# Chapter 1

# Introduction

## 1.1 Background

### 1.1.1 Website Generation with Language Models

Agents built on Large Language Models (LLMs) can now turn a natural-language request into a running website [19]. A user describes the pages and features they need, for example a booking form, a product list and a contact page, and the agent writes a complete project, installs its dependencies and starts it. Benchmarks such as WebGen-Bench evaluate this ability end to end: the generated project is started, and a Vision-Language Model (VLM) tester clicks through the site to check whether each requested feature actually works [19, 11]. This is a demanding test. A website that looks plausible in a screenshot but has a form that does nothing, or a page that cannot be reached from the navigation, fails the corresponding check.

The strongest results on such benchmarks come from large proprietary models. Small open models are attractive for a different reason: they can be run locally, at low cost and without sending data to a third party. WebGen-LM-7B, the 7-billion-parameter generator released with WebGen-Bench, was fine-tuned specifically for this task, yet it reaches only 28.4% accuracy under the official protocol [19]. Improving small generators without retraining them is therefore a practical problem as well as a research one.

### 1.1.2 Every Task Is Solved in Isolation

In WebGen-Bench, as in most code-generation benchmarks, every task is solved independently. The agent starts each task with the same weights and the same empty context. An agent that produced a blank page on one task because it imported a package that was

not installed will make the same mistake on the next task, and on the one after that. In our reproduction of the baseline (Section 4.2), almost a third of the 101 tasks received a score of zero, and many of these failures had the same few causes: an undeclared dependency, a missing import, or a template that does not compile.

A human developer who has broken a build once avoids the same mistake the next time; these failures are exactly what experience should remove.

### 1.1.3 Long-Term Memory for Agents

Recent work on agent memory gives LLMs this ability: after each task the agent reflects on its trajectory, stores lessons or procedures, and retrieves them for later tasks [36, 31, 21, 32, 5]. Almost all of these systems use large models and store memory as natural-language insight. Small models, however, follow many instructions less reliably [10, 13, 17], are distracted by irrelevant context [24] and degrade as prompts grow [8, 18]. A memory that adds a paragraph of lessons to every prompt may help a large model and hurt a small one; which *form* of memory a small model can use is an open question.

## 1.2 Problem Statement

This dissertation asks whether a system built around a small website generator can get better with experience, without any change to the generator's weights. We use *long-term memory* in an operational sense: information is written after a task, persists across tasks, is retrieved under a stated condition, and changes the prompt of later tasks. *Learning* accordingly means a persistent, outcome-dependent change in the system's future behaviour, held in an external memory; the generator's parameters never change, and the memory is maintained by a separate, larger memory model (Chapter 3). We study the question on WebGen-Bench with WebGen-LM-7B. The generator, the agent scaffold (Bolt.diy) and the official evaluation protocol are kept fixed; the only thing that changes between arms is the text added to the generator's prompt. The memory is built only from the agent's own earlier tasks and their evaluation results; the hidden evaluation of the current task is never used, so the comparison with the no-memory baseline is fair.

Two properties make this setting a demanding test of long-term memory. First, tasks arrive in a sequence and the memory available to a task depends on which tasks came before it, so the result may depend on task order. Second, under the official protocol

the generator has a single attempt per task and no repair loop, so the memory must prevent an error rather than help fix it afterwards.

## 1.3 Research Questions

*RQ1.* Can a cross-task long-term memory, built only from the agent's own earlier tasks, improve the functional accuracy of a small website generator under the official WebGen-Bench protocol?

*RQ2.* What form of memory does a small generator actually use: textual lessons, structured requirement checklists, or code examples?

*RQ3.* Does the benefit grow as experience accumulates, and is it robust to the order in which tasks are presented?

*RQ1* asks whether there is an effect at all. *RQ2* asks what produces it, which matters for designing memory for small models in other domains. *RQ3* asks whether the effect behaves like learning from experience, rather than like a fixed prompt change that would help equally on every task, and whether it survives a change of task order. The fixed-prefix ablation (Section 5.5) addresses the first part directly.

## 1.4 Scope

The study is deliberately narrow so that its comparisons are clean.

- **One benchmark, one generator, one scaffold.** All experiments use the 101 tasks of WebGen-Bench and Bolt.diy; the main generator is WebGen-LM-7B, and WebGen-LM-32B from the same family is used to test whether the findings depend on model size (Section 5.6). Other agents such as OpenHands or Aider are not studied, because changing the agent also changes how memory reaches the model.
- **Official protocol, no repair.** Each website is generated once and evaluated as it is, as in the original benchmark. We do not add build checks or repair loops, which would make the results incomparable with the published baseline.
- **No retraining.** Memory only changes the prompt. Fine-tuning on stored experience is outside the scope.
- **Self-hosted models only.** Generation, evaluation and memory processing all run on a university GPU cluster; no commercial API is used.

## 1.5 Contributions

1. A faithful, fully self-hosted reproduction of the WebGen-Bench No Memory baseline for WebGen-LM-7B (28.5% vs 28.4% reported), together with a failure analysis showing that a large share of zero scores are blank pages caused by basic code errors (Chapter 3 and section 4.2).

2. A long-term memory pipeline for website generation with three components, a requirement checklist, a code pool of fragments copied verbatim from the chain's own compiled websites, and memory hygiene, each motivated by a measured failure of the previous version (Chapter 4).

3. Evidence that, for this small generator, short code fragments were followed where textual advice was not, and that a text-dominated memory fell behind No Memory as generic lessons accumulated. On the original task order the code-based memory (v7) raises accuracy from 28.5% to 40.0% (paired gain 11.5 points, 95% Confidence Interval (CI) 6.0–16.9), and an ablation attributes most of this gain to the memory rather than to the checklist (Chapter 5). [TODO: Status: v9.1 on the manifest order has finished generation and is being scored (due 10-16). Replace this item with the v9.1 result, the final method, and describe v7 as an intermediate step.]

4. A task-order study of cross-task memory, which shows that a single memory entry can turn into a failure mode under a different order, and that replacing whole-file examples with verified fragments removes it (Sections 5.7 and 5.7.1).

5. An experimental design for cross-task memory that controls for time drift by running the two arms as alternating pairs, reports task-paired confidence intervals, and is explicit about the limits of single-run results (Chapters 5 and 6).

## 1.6 Organisation of Thesis

**Chapter 2: Background and Related Work**
Website-generation benchmarks, memory for LLM agents, and why small models are different.

**Chapter 3: Experimental Setup**
The benchmark, models, cluster, protocol and metrics, and the reproduced baseline.

**Chapter 4: Method**

The failure analysis that started the work, the design iterations, and the final pipeline.

**Chapter 5: Results**

Main comparison, learning curves, error analysis, ablation and task-order experiments.

**Chapter 6: Discussion**

Interpretation, limitations, threats to validity and ethical considerations.

**Chapter 7: Conclusion**

Answers to the research questions and future work.

# Chapter 2

# Background and Related Work

This chapter places the work in four bodies of literature. Section 2.1 describes how website generation is evaluated and why WebGen-Bench and WebGen-LM-7B were chosen. Section 2.2 reviews memory for LLM agents, with attention to the *form* in which experience is stored. Section 2.3 summarises evidence that small models respond differently to added prompt text, and Section 2.4 separates our use of reflection from self-correction on the same task. Section 2.5 states the gap that this dissertation addresses.

## 2.1 Website Generation Benchmarks

Code-generation benchmarks have moved from single functions towards whole applications. A website is a demanding target: it combines several files, a build configuration, a user interface and interaction logic, and a request in natural language usually lists several pages and features at once. Evaluating such output by reading the code is unreliable, because a project can look complete and still fail to start, or start and render a blank page. Recent benchmarks therefore start the generated project and test it through its interface. Related benchmarks test agents that operate existing websites, such as Mind2Web and WebArena [6, 40], generate a single page from a screenshot and compare it visually, as in Design2Code [27], or change a code repository and run its tests, as in SWE-bench [14].

WebGen-Bench [19] contains 101 website requirements from 20 application domains, each paired with a set of functional test cases (647 in total). A generated project is installed and started, and a Graphical User Interface (GUI) agent based on WebVoyager [11] executes each test case and returns *Pass*, *Partial* or *Fail*. A separate VLM grades the appearance of the landing page from 1 to 5. The authors also release

WebGen-LM, generators fine-tuned from Qwen2.5-Coder [12] on agent trajectories; with the Bolt.diy scaffold, WebGen-LM-7B reaches 28.4% accuracy and an appearance score of 2.5. Larger general models such as Qwen2.5-Coder-32B score lower (9.5%), which makes the 7B model a good subject: it is neither at the floor nor at the ceiling of the benchmark.

WebGen-Bench evaluates generators inside agent scaffolds rather than as bare models. In the Bolt.diy scaffold used here, the model writes the project as a sequence of file actions in its reply, the scaffold applies them to a template project, installs dependencies and starts a development server. The official protocol gives the generator a single attempt: there is no build check and no repair loop, and a project that does not start receives zero for all of its test cases. This is important for our study, because it means memory cannot help by fixing errors after they happen; it can only change what the generator writes in the first place.

The benchmark paper reports each configuration as a single run without error bars; its NeurIPS checklist states that error bars were not reported because it would be too computationally expensive [19]. The limitations it lists itself are that generation covers only TypeScript, JavaScript, CSS and HTML, and that only supervised fine-tuning was used to improve open models. Because we compare two arms whose difference may be small relative to run-to-run noise, we repeat runs and report paired confidence intervals (Section 5.7).

WebGen-Bench is a static benchmark. It does not define any interaction between tasks, so neither memory across tasks nor the order of tasks is part of its protocol. Introducing memory makes the order a treatment variable, which we handle explicitly in Section 5.7.

## 2.2 Memory for LLM Agents

An agent memory has three parts: what is written after a task (raw trajectories, insights, procedures or code), how it is maintained (added, merged, updated or forgotten), and how it is retrieved and presented to the model for a new task. Most work varies the first two and assumes the consuming model can use whatever is retrieved. The paragraphs below follow the same division and end with the question that matters most here: which form of memory transfers to a new task.

**Experience and insight memory.** ExpeL [36] lets an agent collect trajectories, extract natural-language insights, and maintain them with add, edit, upvote and downvote

operations. ReasoningBank [21] distils reusable reasoning strategies from both successful and failed trajectories. Generative Agents [23] rank memories by recency, importance and relevance. Reflexion [25] stores verbal self-reflections after a failed attempt, but reuses them on retries of the same task rather than on new tasks.

**Skill and code memory.** Voyager [31] stores executable skills as code and retrieves them by description; Agent Workflow Memory [32] stores reusable workflows. Metis [5] promotes recurring plans to validated tools that the agent executes; with a GPT-4o executor and memory built only from earlier tasks, its code memory reaches 53.3% task completion against 73.3% for text memory, brittle under task variation. Our fragments are different: they are placed in the prompt to be imitated, not executed.

**Exemplars and evolving context.** Synapse [38] abstracts earlier trajectories before using them as exemplars, Dynamic Cheatsheet [29] keeps strategies and code snippets updated at inference time, and ACE [35] edits a context playbook incrementally to avoid losing detail. Two lessons shaped our design: examples should be stripped of details from their source task, and entries should be versioned rather than rewritten.

**Which form of memory transfers.** Memory Transfer Learning [15] finds, with strong models, that abstract insights transfer best across domains and concrete trajectories can transfer negatively. Neither it nor Metis tests small models; our results suggest that the preferred form depends on the model that must use the memory.

**Cross-task reuse on WebGen-Bench.** Super Library Agent [28] also builds WebGen-Bench applications in sequence, sharing a library of components. It targets maintainability: on 24 single-topic tasks with a strong backbone, accuracy changes insignificantly (76.04% to 77.21%) while code size falls. It stores no lessons from failures and does not vary task order; we study correctness with a small generator on all 101 tasks, with order as a variable.

**Memory maintenance.** Mem0 [4] decides at write time whether a new memory should be added, should update an old one, or is redundant. Memento [39] learns the utility of stored cases from their outcomes. MemGPT [22] manages several tiers of memory for long interactions. Our hygiene component borrows these ideas: deduplication at storage time and a utility score for each memory.

**Task order in continual agent evaluation.** Benchmarks for agents that learn across tasks report that the order and composition of the task stream affect results. Evo-Memory [33] finds that baselines vary substantially between easy-to-hard and hard-to-easy sequences, and AgentCL [26] shows that memory methods can only be distinguished on streams that expose opportunities for reuse. We therefore compare orders in which tasks of the same domain are adjacent with orders in which they are mixed (Section 5.7).

## 2.3 Small Models, Long Prompts and Instruction Following

Several results suggest that adding text to a prompt is not free. Models become less likely to satisfy all instructions as their number grows [10, 13]; information in the middle of a long context is used less than at its start or end [18]; irrelevant context distracts models [24]; and longer inputs alone can reduce performance even with perfect retrieval [8]. Models can be trained to prioritise instructions by source [30], which motivates keeping the user requirement explicitly above any memory content. Small models also struggle to benefit from long reasoning chains [17], which is consistent with our negative result for a short planning step (Section 4.3).

These findings were obtained mostly on reasoning and question-answering tasks, but they point to a specific risk for memory. A memory that grows with experience adds more and more text to every prompt. For a large model this may be harmless; for a small one it may bury the user's requirement, introduce instructions that conflict with it, or present advice the model cannot turn into code. Whether a small generator can use memory at all, and in which form, therefore cannot be assumed from results obtained with large models.

## 2.4 Self-Correction and Debugging

Reflection on failed attempts is related to self-correction methods such as Self-Refine [20], Self-Debug [3] and Chain-of-Verification [7]. These methods repair the *same* task with feedback. We do not allow this in the formal comparison, because repairing a benchmark task using its hidden evaluation results would leak the answer. Reflection is used only to write memory for *later* tasks. The investigation steps we ask the reflection model to follow are modelled on systematic debugging [34].

## 2.5 Research Gap

Table 2.1 compares the closest studies along the dimensions that matter for this dissertation. Three gaps follow from it. First, cross-task memory has been studied almost only with large agents; whether a small generator benefits, and from which form of memory, is untested. Second, on WebGen-Bench the only cross-task study targets maintainability with a strong backbone on single-topic suites, and reports no significant change in functional accuracy. Third, memory results are usually reported for a single task order, although the memory available to a task depends on the tasks before it. This dissertation addresses all three: it measures the effect of memory on functional accuracy for a 7B generator under the official protocol, compares textual, checklist and code forms of memory, and repeats the comparison under different task orders with task-paired confidence intervals.

Table 2.1: Closest related studies. Small: the model that uses the memory has 7B parameters or fewer. Websites: evaluated on website-generation tasks. Order: results compared across task orders.

[TODO: Planned: add columns for memory form, source, update rule, retrieval and consumer, and mark "not evaluated" separately from a shown limitation.]

| Study | Memory form | Small | Websites | Order |
|---|---|---|---|---|
| ExpeL [36] | textual insights | no | no | no |
| ReasoningBank [21] | reasoning strategies | no | no | no |
| Voyager [31] | executable skills | no | no | no |
| Metis [5] | text + executed code tools | no | no | no |
| Memory Transfer Learning [15] | four representations | no | no | no |
| Super Library Agent [28] | shared component library | no | 24 tasks | no |
| Evo-Memory [33] | benchmark of memory methods | – | no | yes |
| **This work** | checklist + code fragments + text | **7B** | **101 tasks** | **yes** |

# Chapter 3

# Experimental Setup

## 3.1 Benchmark and Protocol

We use all 101 WebGen-Bench tasks. They come from 20 application domains of unequal size, from eight tasks each (company brochures, learning platforms, discussion forums, e-commerce) down to two (personal blogs, streaming, email clients). The benchmark's manifest order groups tasks into domain blocks; Section 5.7 adds shuffled orders. We follow the official protocol: the generator produces the project once, there is no repair loop, and a project is passed to evaluation as soon as its dependencies install and the development server starts. Test cases without an evaluation record count as failed, as in the official scoring script. An earlier phase of this project used a stricter build gate (`vue-tsc`) and a more generous repair loop; those results are not comparable with the paper and are not reported as main results.

Figure 3.1 shows one task in the pipeline. The memory module is the only component that differs between arms; the generator, the scaffold, the tester and the scoring are shared.

## 3.2 How One Task Is Generated and Scored

This section follows one task through the unmodified system, so that it is clear what the memory later changes. Task 000006 asks for a solar dashboard:

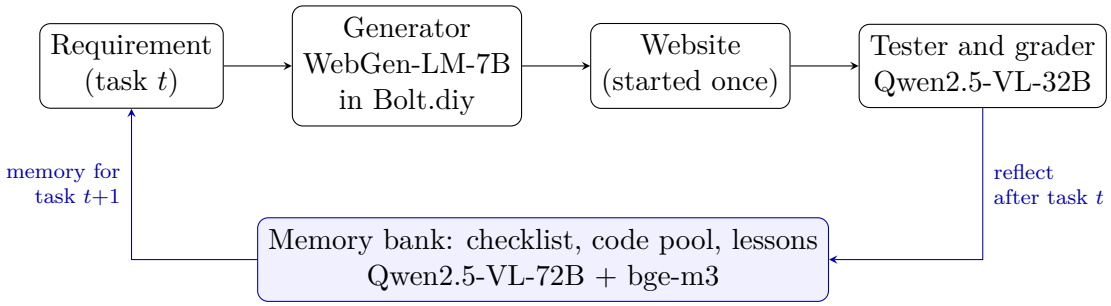



Figure 3.1: One task in the pipeline. Black: the official WebGen-Bench setup (No Memory). Blue: the memory module, which reflects after each task and adds text to the prompt of later tasks only.

```
Please implement a solar dashboard to display real-time and historical solar
data. The dashboard should have functionalities for real-time data display,
historical data query, and chart analysis. [...] Configure the background
    color
to honeydew, with components using dark olive green.
```

**Generation.** Under No Memory this text is the entire prompt. Bolt.diy is an open-source browser-based coding agent: it opens a starter project (for Vue, a Vite + Vue + TypeScript template whose `package.json` declares only `vue`), sends the requirement to the generator, and writes every file of the model's reply into the project. It then installs the dependencies and starts the development server. The generator has one reply; nothing in the official protocol inspects or repairs the code.

**Functional testing.** Each task comes with hidden test cases, on average six. For task 000006 the first two are "Verify the display of real-time solar power generation data" and "Test the display of real-time power consumption data", each with an expected result. A WebVoyager-style GUI agent [11] receives one test case at a time. At each step it sees a screenshot in which every interactive element carries a numbered box (Figure 3.2b), chooses an action such as clicking element 1 or typing into a field, and finally answers YES, PARTIAL or NO. The agent has a fixed step budget; if it runs out before reaching the target, the case counts as failed.

**Appearance grading.** A vision model grades a full-page screenshot from 1 to 5 for layout, colour and completeness. Both the tester and the grader are models acting as judges, which are known to have biases [37]; we therefore keep their raw transcripts and, in Section 6.4, check a sample of their verdicts by hand.

Figure 3.2 shows what the tester saw for this task in two runs. Under No Memory the model imported the chart library `vue-chartjs`, which the template does not declare; Vite cannot resolve the import, the page shows only the error overlay, and all test cases fail. With the final memory the generator built the three requested views without the undeclared package, and all test cases passed.




(a) No Memory: score 0.

(b) Final memory (v9.1): score 1.0.

Figure 3.2: The first screenshot the tester received for task 000006 (seed-11 order). (a) An undeclared package stops the whole site. (b) The numbered dashed boxes are drawn by the tester to name the elements it can click; they are not part of the website.

## 3.3 Models and Roles

All models are self-hosted on the university CS ML cluster (two NVIDIA H200 GPUs per job, served with vLLM [16]). No commercial LLM service is called. Table 3.1 lists the roles.

Table 3.1: Model roles. The appearance grader is the only deviation from the paper, which uses GPT-4o.

| Role | Model |
|---|---|
| Generator | WebGen-LM-7B (BF16), Bolt.diy scaffold, single generation |
| Functional tester | Qwen2.5-VL-32B-Instruct [1] (WebVoyager-style GUI agent) |
| Appearance grader | Qwen2.5-VL-32B-Instruct (paper: GPT-4o) |
| Memory / reflection | Qwen2.5-VL-72B-Instruct-AWQ |
| Embeddings | bge-m3 [2] |

**The generator.** WebGen-LM is a family of website-generation models (7B, 14B and 32B) released with WebGen-Bench [19]. Each is Qwen2.5-Coder-Instruct [12] of the same size, fine-tuned with supervised learning only. The training data come from WebGen-Instruct, 6,667 website instructions decontaminated against the benchmark: the authors ran DeepSeek-V3 in Bolt.diy on 2,000 of them and kept the 600 trajectories whose websites scored at least 3 for appearance. WebGen-LM-7B therefore already knows the Bolt.diy output format, which makes it the natural small baseline for this benchmark: it is not a general model prompted to build websites, but a model distilled for exactly this scaffold.

The memory model is deliberately different from the evaluator, so that reflection cannot copy the evaluator's grading behaviour. vLLM applies the model repository's generation configuration by default; WebGen-LM-7B ships with a repetition penalty that Bolt.diy never sends, so we disable it for the generator to match the paper's setting.

## 3.4 Metrics

For each task with $n$ test cases, of which $p$ pass and $q$ are partial,

$$\text{accuracy} = \frac{p + 0.5\,q}{n}. \tag{3.1}$$

We report accuracy over all test cases, the mean appearance score (1–5), and the number of *zero-score* tasks. Because test cases within a task are not independent, comparisons are paired by task rather than by test case: for two arms we report the mean per-task difference in accuracy with a 95% percentile bootstrap CI obtained by resampling tasks [9], together with the numbers of tasks won, lost and tied. This interval is conditional on the observed runs: it reflects which tasks happened to be in the benchmark, but not the variation between repeated runs of the same memory chain (about five points, Section 5.7.1), nor the dependence between tasks that a memory creates. We therefore say that an interval "excludes zero" rather than calling a gain significant. Accuracy pools test cases (micro); the learning curves average tasks (macro). [TODO: Planned (existing data, no new run): add task-macro and domain-macro accuracy and a domain-cluster bootstrap for the final comparison.]

## 3.5 Experimental Design

The memory designs we built fall into three families, shown in Figure 3.3 and described in Chapter 4. *Text memory* (v1–v4) stores lessons written in natural language and adds them to the prompt; its last version, v4, adds the requirement checklist. *Code memory* (v5–v7) keeps the checklist and adds a code pool of examples taken from the chain's own working websites. The *final* family (v8–v9.1) changes only how the code pool selects and updates its examples. One version of each family was run on all 101 tasks, v4, v7 and v9.1, and in the rest of this dissertation the three names refer to these runs. Versions that were stopped part-way or never run on all tasks are described only in the design history (Section 4.3).

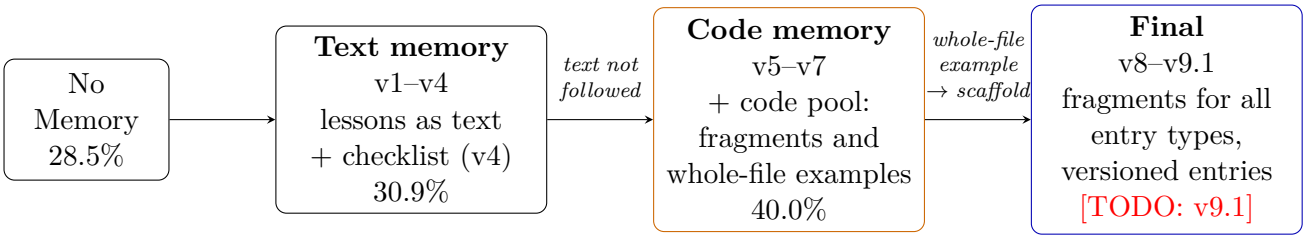



Figure 3.3: The three families of memory. Each arrow names the failure that led to the next family. Accuracy is on the manifest order (101 tasks, one run).

Table 3.2 lists every arm, the task orders in which it was run and the question it answers. All arms use WebGen-LM-7B with Bolt.diy under the official protocol, cover all 101 tasks and start from an empty memory; each arm is run once per order. The task orders are defined in Section 5.7.

Memory never uses information from the current task itself (leave-one-task-out), and evaluation results are never fed back into the same task.

## 3.6 Reproducing the Baseline

No Memory on all 101 tasks reaches 28.5% accuracy (paper: 28.4%) with a mean appearance score of 2.35 (paper: 2.5 with GPT-4o) and 30 zero-score tasks. The close match in accuracy gives us confidence that the self-hosted pipeline is faithful to the official one.

Table 3.2: Experimental design. Memory content is what is added to the generator's prompt; the families are those of Figure 3.3.

| Arm | Memory content | Task orders | Question |
|---|---|---|---|
| No Memory | none (requirement only) | manifest, seeds 11, 29, 47, interleaved | Does the baseline reproduce the paper, and how much does it vary between runs? |
| Text memory (v4) | checklist with textual lessons | manifest | Does textual memory help a small generator? |
| Checklist only | checklist; memory neither read nor written | manifest | How much does decomposing the requirement add? |
| Code memory (v7) | checklist, lessons and code examples | manifest, seeds 11, 29 | Does code memory help, and does it survive a change of order? |
| Fixed code prefix | checklist and the final code memories from task 1; no memory | manifest | Is the gain learning, or just code in the prompt? |
| Final (v9.1) | as v7, code examples are verified fragments only | manifest, seeds 11, 29, 47, interleaved | Does the final method hold across task orders? |
| 32B: No Memory and Final (v9.1) | as above, generator WebGen-LM-32B | manifest | Do the findings hold for a stronger generator? |

## 3.7 Reproducibility and Constraints

Every run starts from an empty memory and stores its configuration file, task order, prompts, retrieved and injected memory, raw evaluator transcripts and scores; the key files of every run are archived in the project repository. The generator is decoded deterministically, so differences between arms come from the prompt rather than from sampling, although the GUI tester is not deterministic.

**Hardware.** All experiments ran on one node of the School of Computer Science CS ML cluster (`foscsmlprd03`, seven NVIDIA H200 GPUs with 141 GB of memory each, Slurm scheduler), with at most two GPUs per user. A single long-running job held two GPUs and served every model with vLLM; all arms and task orders shared these servers (Table 3.3). Two tasks were processed in parallel, while evaluation was serialised by a lock, because the evaluator restarts the website process manager. From the first job on

5 October 2026 to the end of the 7B runs, the project used about [TODO: update] H200 GPU-hours.

Table 3.3: Placement of the self-hosted models on the two H200 GPUs (memory reserved by vLLM).

| GPU | Model (role) | Memory (GB) |
|---|---|---|
| GPU 0 | Qwen2.5-VL-32B-Instruct, BF16 (evaluator) | 83 |
| | WebGen-LM-7B, BF16 (generator) | 26 |
| GPU 1 | Qwen2.5-VL-72B-Instruct-AWQ (memory) | 113 |
| | bge-m3 (embeddings) | 2 |

**Code and data availability.** The code, configuration files and the archived run records (prompts, injected memory, scores and memory databases of every run) are publicly available at https://github.com/qiaopan/WebGen-Bench; the run records are in `archive/phase2-webgen-2026-10/`.

The main constraint is compute. A task takes about twelve minutes with memory and six without, so one 101-task run takes about a day on our two GPUs. As in the original benchmark, each configuration is therefore run once per task order. We compensate by pairing comparisons by task, by repeating the No Memory baseline in three orders to measure its run-to-run variation, and by repeating the memory method across orders rather than within one.

# Chapter 4

# Method

## 4.1 Problem Formulation

A run processes the tasks $t_1, \ldots, t_n$ ($n = 101$) in a fixed order. Task $t_i$ has a requirement $r_i$ and hidden test cases $c_{i,1}, \ldots, c_{i,m_i}$. A generator $G$ with fixed weights maps a prompt $x_i$ to a project $y_i = G(x_i)$, which the evaluator $E$ scores as $a_i = E(y_i, c_{i,\cdot}) \in [0, 1]$ with the accuracy defined in Chapter 3. Under No Memory, $x_i = r_i$. With memory, the prompt is

$$x_i = r_i \oplus f(r_i, M_{i-1}), \qquad M_i = u\big(M_{i-1}, r_i, y_i, \tau_i, a_i\big), \tag{4.1}$$

where $M_0 = \emptyset$, $f$ selects and formats memory for the current requirement, $\oplus$ appends text after the requirement, $\tau_i$ is the evaluator transcript, and $u$ reflects on the finished task and updates the memory. Two constraints make the comparison fair: $f$ only reads $M_{i-1}$, which is built from earlier tasks, so the hidden evaluation of $t_i$ never reaches its own prompt; and $G$, $E$ and the protocol are identical across arms. The question is whether the mean of $a_i$ is higher with memory than with $x_i = r_i$, and which design of $f$ and $u$ achieves it. Algorithm 1 gives the final design.

[TODO: Planned: a pipeline figure for v9.1 that separates the read path (checklist, lesson retrieval, code pool, prompt) from the write path after evaluation (reflection, deduplication, code facts, versioned entries), and shows that the current task's evaluation never reaches its own prompt.]

## 4.2 Starting Point: Why No Memory Fails

We began by inspecting the code that No Memory actually produced, rather than its scores. Two findings shaped everything that followed.

**Algorithm 1** Cross-task memory for one run (final version).

$M \leftarrow \emptyset$
**for** $i = 1, \ldots, n$ **do**
  $L \leftarrow \textsc{Checklist}(r_i)$ ▷ memory model splits the requirement
  attach to $L$ the lessons in $M$ judged to apply to $r_i$
  $P \leftarrow$ up to three code-pool fragments whose trigger holds
  $x_i \leftarrow r_i \oplus L \oplus P$; $y_i \leftarrow G(x_i)$; $a_i \leftarrow E(y_i)$
  reflect on $(y_i, \tau_i, a_i)$; add new lessons to $M$ after deduplication
  record code facts of $y_i$; create or version code-pool entries from earlier sites with $a \geq 0.5$
**end for**

**Blank pages from basic errors.** Bolt.diy's Vue template declares only `vue` as a dependency, and WebGen-LM-7B never edits `package.json` (0 of 81 inspected responses). When it imports `vue-router` or uses a Vue function such as `ref` without importing it, the whole site renders as a blank page and every test case fails. In an early Long Memory run, 26 of 74 tasks were blank, 19 of them because of an undeclared `vue-router` import. Many zero-score tasks therefore fail before their functional difficulty can be assessed: the site does not render because of a basic error that the model repeats.

**Text is not followed.** In a diagnostic on nine tasks that had produced blank pages, we added a single sentence to the prompt, phrased in several ways ("add vue-router to package.json", "do not use vue-router", "put pages in App.vue with `ref` and `v-if`", or the same sentence at both the start and the end). None recovered more than one task in three. A ten-line code example of the same idea recovered seven of nine. This matches the observation that small models follow long textual guidance poorly [10, 17] and suggested that memory should be delivered as code.

## 4.3 Design History

The three families of Figure 3.3 were reached through the attempts in Table 4.1; unlike Table 3.2, which lists the arms compared, the table records how the method evolved. Versions are numbered in build order. v1 and v2 are early text-memory designs, v5 was a local prototype, and v8 and v9 were stopped part-way; only v4, v7 and v9.1 ran on all 101 tasks.

**From text to code.** The first memory followed ExpeL-style reflection: after each task the memory model wrote cases (trigger, observation, cause, preventative action) with

evidence, retrieved later as guidance. Reflection rules teach a method of investigation, never answers, so that the memory is learned rather than designed. After a scaffold bug was fixed (Bolt.diy treated a line break as "send" and dropped the rest of the prompt), long guidance made the model build only the home page, and compressing it dropped features. The checklist of v4 keeps every requested feature visible and attaches lessons to the item they concern; v4 beat No Memory on the first 30 tasks (39.3% vs 26.0%) but not on the last 71 (27.8% vs 29.5%). The code pool of v6 replaced text by code, but its one-line example wrapped in `<script setup>` tags looked like a finished empty component and produced untouched scaffolds; v7 therefore admits, for missing-import entries, only verbatim fragments without tag lines that include a line using the imported function. Missing-package entries in v7 still showed a complete `App.vue` of an earlier site, because their example had to demonstrate page switching without the missing package. In the manifest order this caused two scaffolds; under shuffled orders it caused many (Figure 4.1); v8 to v9.1 change only the code pool to fix this (Section 5.7.1). A planning step before generation was rejected: in five of five tasks the generator wrote only the plan and lost Bolt.diy's file format.

```
CODE FROM EARLIER WEBSITES THAT WORKED:
Missing package: imports a package that
package.json does not declare. [...] Code
that works:
<script setup lang="ts">
import { ref } from 'vue'
import FoodDonationPage from
  './pages/FoodDonationPage.vue'
[... 3 more page imports ...]
const activeTab = ref('donation')
</script>
<template>
  <div class="app-container">
    <header>
      <h1>Food Distribution System</h1>
      [... 30 lines: nav, four pages ...]
</template>
```




Figure 4.1: Code memory (v7) on task 000041, an online trading site (seed-11 order). Left: part of the code-pool entry in the prompt, a complete `App.vue` from an earlier food-donation site. Right: the generator returned the untouched Vite starter project, and all five test cases failed. The final version shows only verified lines of such a file, without the `<script>` and `<template>` tags (Figure 4.2).

## 4.4 Final Pipeline (v9.1)

For each task, in order:

1. **Requirement checklist.** The memory model splits the requirement into items (Section 4.4.1).
2. **Retrieve lessons.** Functional lessons are retrieved by similarity and judged for applicability by the memory model; only hard matches are attached to checklist items. A small common pool adds site-wide lessons that contain a concrete code construct.
3. **Code pool.** At most three verified fragments whose trigger condition holds (Section 4.4.2).
4. **Generate and evaluate.** One generation by WebGen-LM-7B; official functional and appearance evaluation.
5. **Reflect and store.** The memory model reflects on the evaluated task; new lessons are deduplicated against existing ones, utilities are updated, and code facts are computed for the code pool.

The products of step 5 are read by steps 2 and 3 of later tasks; this is the only path through which tasks influence each other. Figure 4.2 shows the resulting prompt for one task; under No Memory the prompt is the first paragraph only.

```
Please implement an online trading website for bidding and trading. [...] Set the
background color to lemon chiffon and the component color to chocolate.

FEATURE CHECKLIST (the requirement above, item by item; build every item):
1. Please implement an online trading website for bidding and trading.
3. allowing users to browse trading opportunities,
   Note from earlier websites: For the trading opportunities: use `v-for` to
   display the list with current status.
[... items 2, 4-8 ...]

CODE FROM EARLIER WEBSITES THAT WORKED:
Component fails: ref() not imported. Earlier websites broke here (a component
calls ref() without importing it from 'vue' (that page fails when opened),
19 tasks). Code that works:
    import { ref, onMounted } from 'vue'
    const wheelRef = ref<HTMLElement | null>(null)
Vue compile error: v-model on prop. [...] (4 tasks). Code that works:
    <SearchBar :modelValue="searchQuery"
               @update:modelValue="emit('update:searchQuery', $event)" />
[... one more entry ...]

Build every checklist item; the notes only warn about mistakes earlier websites made.
```

Figure 4.2: The prompt the final version (v9.1) gave the generator for task 000041 (seed-11 order), shortened. The requirement comes first and stays binding; the checklist repeats it item by item with lessons attached; the code pool adds verified fragments with the number of earlier tasks that failed in the same way.

### 4.4.1 Requirement Checklist

The original requirement remains at the top of the prompt and is declared binding. The checklist below it repeats every requested feature, so that memory content cannot crowd a feature out of the model's attention [18, 30]. Lessons are attached to the item they concern rather than listed separately, which avoids a separate list being read as the full scope of the task.

### 4.4.2 Code Pool

The code pool is computed without an LLM for detection:

- **Problem facts.** After each task, code analysis of the generated site records undeclared imports (`import:<package>`), Vue functions used without import (`vue-api`), and compile errors obtained by compiling each Single-File Component (Vue) (SFC) with `@vue/compiler-sfc` exactly as Vite would.
- **Trigger.** A problem key becomes active after it has appeared in two different tasks, following Metis's rule of promoting a procedure only after it recurs [5]. Whole-site failures that code detects objectively (an undeclared package, a compile error) become active after one occurrence. Missing-import and compile-error entries apply to every Vue site; other entries are applied only if the memory model judges their condition to hold for the requirement.
- **Source.** The example comes from the chain's own earlier websites that scored at least 0.5 and compile. The threshold of 0.5 is a design choice (at least half the checks passed), not a tuned value. Successful sites supply *how* to write it; failed sites decide *when* it is needed [21].
- **Fragment selection.** The memory model picks the key lines; code then verifies that the lines are verbatim from the source file, contain at most 15 lines, contain no `<script>`/`<template>` tag lines, and include the import of the most frequently missed function plus a line that calls it. If the model fails twice, a rule-based extraction is used. An entry is updated by adding a new version only when the new example covers strictly more of the needed functions; the old version is kept for provenance [35].
- **Slots.** At most three entries per task, at most one of them for a missing package.

*Verified* in this dissertation means only that a fragment is verbatim from a source file that compiles and comes from a site with score $\geq 0.5$; it does not mean that the fragment is correct for the new task. Because the triggers are general, the code pool behaves partly like a learned prompt prefix: once active, many later tasks receive the same fragment. The fixed-prefix ablation (Section 5.5) separates this from learning. [TODO: Planned (appendix, no new run): retrieval and storage details: entry fields, embedding input, number of candidates, similarity threshold, injection budget, mean injected tokens per task, and the applicability and reflection prompts.]

### 4.4.3 Memory Hygiene

- **Concrete only.** Common-pool entries without a code construct (for example `@click`, `v-if`, an `import` line) are dropped. Rewriting may generalise the *condition* under which a lesson applies but not the *action*.
- **Deduplication at storage time.** A new lesson is compared with similar stored lessons; if the memory model judges it to share the same root cause and fix, it is recorded as support for the old lesson instead of a new entry [4].
- **Utility and retirement.** For each lesson we count injections $u$ and recurrences $r$ (the same problem reappearing after injection), and compute utility $(u - r + 1)/(u + 2)$ [39, 36]. Ranking uses support $\times$ severity $\times$ utility; a lesson injected at least five times with utility below 0.3 is no longer injected. Utility is a heuristic: a task after injection that does not repeat the problem may not have needed the lesson, and a lesson used on hard tasks can be penalised. The constants (two occurrences, five injections, 0.3) are design choices that were not tuned.
- **No hand-written conclusions.** Instructions to the reflection model describe how to investigate (follow the code path, trace styles from element to global stylesheet, list candidate causes and rule them out) but never what the answer is.

Table 4.1: Design iterations. Each row addresses a failure observed in the previous row.

| Attempt (version) | Problem addressed | Inspired by | Outcome |
|---|---|---|---|
| Three-layer memory (v1–v2) | Traceable lessons | [36, 31, 21] | Abstract experiences too vague; kept concrete cases only |
| Guidance compression | Injected text 5.7× the requirement; only the home page was built | [10, 18] | One task 0.25 → 0.83, but compression dropped required features |
| Step-wise reflection | Guesses stored as lessons | [34, 3, 7] | Better evidence; high run-to-run variance |
| Dedup + utility (v3) | 34 near-identical lessons filled the pool | [36, 4, 39] | Less repetition; blank pages unchanged |
| Requirement checklist (v4) | Requirement drowned by guidance | [30, 18] | Strong early, worse than No Memory later |
| Code pool (v5 prototype, v6) | Text not followed; generic entries | [5, 31, 32] | A one-line example caused untouched scaffolds |
| Fragments for missing imports (v7) | Example read as a finished file | – | Best of the first round; missing-package entries still whole files |
| v8 | Whole-file missing-package example caused scaffolds after a change of task order | [32, 38, 35] | All entry types fragments; scaffolds removed; example kept switching versions |
| v9 | Rare whole-site failures learned too late; two slots too few | [5] | Scaffolds stay at zero; v-model memory never applied |
| v9.1 (final) | Applicability condition never matched | – | Final version (Section 5.7.1) |
| Short CoT (rejected) | Missing imports | [17] | 0/5 produced any files |

# Chapter 5

# Results

Each arm is a single run over all 101 tasks. Sections 5.1 to 5.5 use the manifest order; Section 5.7 repeats the comparison in other orders. Accuracy pools all test cases (as in the official script); the learning curves average per-task accuracy, so their end points differ slightly.

## 5.1 Main Comparison

Table 5.1 reports the main comparison in the format of the WebGen-Bench results table [19], with the published WebGen-LM-7B result as the first row. Every arm uses the same generator, scaffold and evaluators; only the memory differs.

Table 5.1: Main results on all 101 tasks in the manifest order (single run per arm). Yes, Partial, No and Start failed are percentages of the 647 test cases; Accuracy = Yes + 0.5× Partial. Appearance is the mean grade over all 101 tasks (1–5; tasks without a working site count as 0). Δ is the mean per-task difference to No Memory with a 95% bootstrap CI. Zero: tasks where every test case fails. Best values in bold.

| Arm | Yes | Partial | No | Start fail. | Acc. | Appear. | Zero | Δ Acc. [95% CI] |
|---|---|---|---|---|---|---|---|---|
| WebGen-LM-7B (paper)[a] | 24.9 | 7.1 | 68.0 | 0.0 | 28.4 | 2.5 | – | – |
| No Memory (ours) | 23.8 | 9.4 | 66.5 | 0.3 | 28.5 | 2.35 | 30 | – |
| Text memory (v4) | 25.2 | 11.4 | 62.6 | 0.8 | 30.9 | 2.57 | 20 | +2.6 [−3.6, 8.7] |
| Checklist only | 26.4 | 10.7 | 61.4 | 1.5 | 31.8 | 2.47 | 18 | +3.4 [−2.2, 8.7] |
| Code memory (v7) | **34.5** | 11.1 | **53.2** | 1.2 | **40.0** | **2.83** | **13** | **+11.5** [6.0, 16.9] |
| Final, verified fragments (v9.1) | [TODO: running] | | | | | | | |

[a] Reported by Lu et al. [19]; appearance graded by GPT-4o, ours by Qwen2.5-VL-32B.

Code memory (v7) raises the share of passed test cases from 23.8% to 34.5% and halves the number of zero-score tasks; per task it beats No Memory on 54 tasks, loses on 24 and ties on 23. The gain over text memory is +8.9 points ($[2.9, 15.1]$; 53 wins, 29 losses, 19 ties). For neither text memory nor the checklist alone does the interval exclude zero. For scale, No Memory itself reaches 28.5%, 27.9% and 28.6% in three independent runs under different orders (Section 5.7); its run-to-run spread of 0.7 points is far smaller than the 11.5-point gain of code memory. A Long Memory chain varies more than No Memory, because its memory diverges from the first task (Section 5.7.1). Appearance in Table 5.1 counts tasks without a working site as 0, so it rises when more sites work; on the 68 tasks where both No Memory and v7 produced a graded site, the means are 3.25 and 3.22. [TODO: Planned (existing data): split the gain into (a) tasks where only the memory arm produced a working site and (b) tasks where both arms worked, and report the paired gain and CI on (b). Do this for the final version once its manifest run is scored.]

## 5.2 Learning Curve

Figure 5.1 plots cumulative and rolling accuracy in execution order. v4 and v7 are similar on the first 30 tasks (39.3% and 39.0%, versus 26.0% for No Memory), when the code pool has few active entries. On the remaining 71 tasks v4 falls below No Memory (27.8% vs 29.5%, 19 zero-score tasks each), while v7 keeps its level (40.4%, 9 zero-score tasks). On the last 35 tasks the gap is 38.3% (v7) vs 26.0% (No Memory) vs 23.2% (v4).

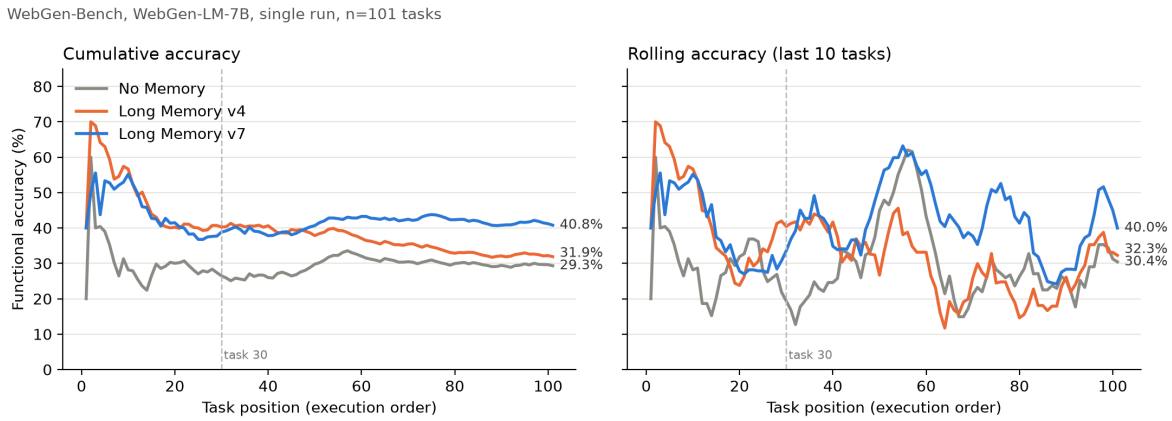



Figure 5.1: Cumulative and rolling-10 accuracy by execution order for No Memory, v4 and v7. [TODO: Planned: mark when each code-pool entry became active, and add the final version.]

Figure 5.2 shows the same data as a paired gap to No Memory. Both memories lead early; between tasks 40 and 60 v4 falls up to 27 points below No Memory while v7 stays near or above zero, and from task 60 onwards v7 stays clearly positive.

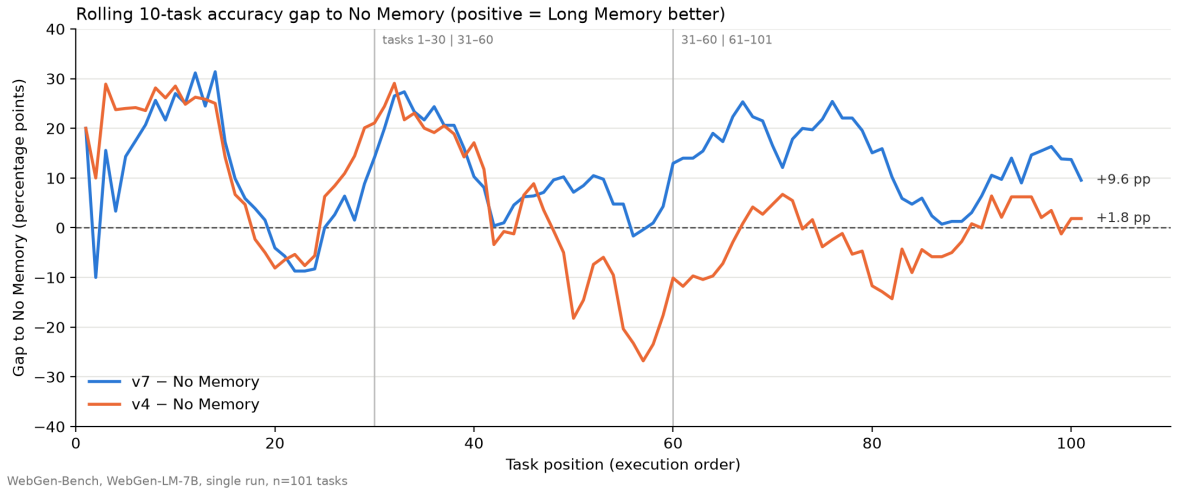



Figure 5.2: Rolling 10-task mean of the per-task accuracy gap to No Memory, by execution order. Positive values mean Long Memory is better.

## 5.3 Why v4 Declines and v7 Improves

Table 5.2 counts what reached the generator's prompt, and Table 5.3 counts code problems in the generated sites, obtained by compiling every site with `@vue/compiler-sfc` 3.5.43. A lesson is counted as *without a code construct* if it contains none of `@click`, `v-if`, `v-for`, `v-model`, `ref`, an `import` or inline code. In v4 such lessons appear in most tasks throughout and a separate "General notes" block becomes more frequent over time, while nothing addresses blank pages. In v7 these lessons are rarer and the code pool supplies a fragment in almost every task after it activates.

These counts are correlational. Disentangling the contribution of each component requires the ablations in Section 5.5.

## 5.4 Error Analysis

Whole-site breaks fall from 26 tasks (No Memory) to 8 (v7), whereas pages with a missing Vue import are about as frequent as without memory (23 vs 22): the code pool prevents the error that blanks the whole site more reliably than the per-page one, which

Table 5.2: Prompt content by segment (tasks 1–30 / 31–60 / 61–101).

| Prompt content | v4 | v7 |
|---|---|---|
| Tasks with a lesson without a code construct | 23 / 25 / 28 | 12 / 4 / 11 |
| Tasks with a "General notes" block | 13 / 16 / 27 | 3 / 0 / 8 |
| Tasks with a code-pool fragment | 0 / 0 / 0 | 25 / 30 / 41 |
| Lessons attached to checklist items | 56 / 53 / 81 | 32 / 16 / 41 |
| of which without a code construct (all tasks) | 102 of 190 | 31 of 89 |

Table 5.3: Generated sites with code problems (number of tasks; 1–30 / 31–60 / 61–101, total in brackets).

| Problem | No Memory | v4 | v7 |
|---|---|---|---|
| Whole-site break[a] | 11 / 5 / 10 (26) | 2 / 5 / 12 (19) | 4 / 1 / 3 (**8**) |
| Missing Vue import on a page | 7 / 7 / 9 (23) | 14 / 10 / 7 (31) | 12 / 4 / 6 (22) |

[a] An undeclared package import or a Vue compile error.

remains a basic error of the 7B generator. Counted per file, 27 of 592 generated Vue components in v7 (4.6%) call a Vue function without importing it, against 27 of 586 (4.6%) for No Memory. The omission occurs in `App.vue` and in child components alike, and is unrelated to whether a component uses `defineProps`/`defineEmits`; in the same file other imports are usually correct. The two arms show the same per-file rate, which is consistent with a slip rather than a misconception that an example could correct, though we have not shown this directly; a site with more components is more likely to contain one.

This slip was not reduced by a better example: a variant that showed both `ref` and `computed` removed it in one of four affected tasks. As the official protocol forbids repairing code after generation, we report it as a limitation of the generator. Ten of v7's 13 zero-score tasks are also zero for No Memory. [TODO: Planned (existing data): list the cause of each remaining zero-score task of the final version (blank page, missing feature, budget exhausted) instead of calling them hard.]

**Appearance.** Restricted to the 68 tasks where both No Memory and v7 produced a working site with an appearance grade, the mean appearance score is 3.25 for No Memory and 3.22 for v7. We observed no meaningful difference on this common subset; because memory changes which tasks enter the subset, this does not show that memory has no

effect on appearance. The three appearance figures in this dissertation use different denominators: 2.35 averages all 101 tasks with failures as 0 (Table 5.1), 3.25 averages tasks with a grade (Table 5.4), and 3.25 vs 3.22 is the paired subset. [TODO: Planned: paired difference with CI on the common subset.]

## 5.5 Ablations: Checklist Only and Fixed Code Prefix

Two ablations separate the contributions of the components, both with the v7 configuration and the manifest order:

- **Checklist only:** v7 with memory reads and writes disabled; each task sees only the requirement checklist. Checklist only − No Memory estimates the effect of requirement decomposition.
- **Fixed code prefix:** checklist only, plus the code-pool entries that v9.1 had learned by the end of its manifest run (Section 4.4.2), injected from the first task with the same selection rule, still without memory. The code is not written by hand; it is the method's own learned content. Because it is taken from the end of a complete run, this control has access to experience that an online learner only gains later; it is a retrospective, oracle-style comparison rather than an online one. Fixed prefix − checklist only estimates the effect of having this code in the prompt, and v9.1 − fixed prefix the effect of accumulating experience and selecting it per task.

If the fixed prefix matches v7, the contribution of the method is better described as automatically discovering effective prompt code from the agent's own failures and successes than as continual accumulation.

Table 5.4 reports the checklist-only ablation over all 101 tasks; the fixed-prefix run is queued after the seed-47 order (expected about 10-18).

Paired by task, checklist only − No Memory is +3.4 points (bootstrap 95% CI $[-2.2, 8.7]$; 51 wins, 27 losses, 23 ties), and v7 − checklist only is +8.1 ($[2.9, 13.2]$; 46/28/27). Requirement decomposition therefore mainly prevents whole-site failures (zero-score tasks fall from 30 to 18) but does not reliably raise accuracy, and its advantage shrinks from +7.1 points in the first 30 tasks to +1.8 afterwards. Memory contributes the larger share, with an interval that excludes zero, and its advantage over the checklist

Table 5.4: Ablation over the 101 tasks in the manifest order (single runs). Accuracy is pooled over checks; appearance averages tasks with a grade.

| | Acc. 1–30 | Acc. 31–101 | Acc. all | Zero-score | Appearance |
|---|---|---|---|---|---|
| No Memory | 26.0 | 29.5 | 28.5 | 30 | 3.25 |
| Checklist only | 33.1 | 31.3 | 31.8 | 18 | 3.23 |
| Fixed code prefix | | | [TODO: running] | | |
| v7 (checklist + memory) | 39.0 | 40.4 | 40.0 | 13 | 3.21 |
| v9.1 (final) | | | [TODO: running] | | |

grows from +5.9 to +9.1 points between the two segments, consistent with the code pool becoming active only after a problem has recurred.

[TODO: Status: fixed-prefix run queued after seed 47. Fill in its row and the differences fixed prefix − checklist only and v9.1 − fixed prefix. Wording depends on the result: if fixed prefix ≈ v9.1, describe the contribution as automatically discovering effective prompt code; if v9.1 is clearly better, as accumulation and per-task selection.]

## 5.6 Generalisation to a Larger Generator

Does the memory framework also improve a larger generator from the same family? We repeat No Memory and the final version with WebGen-LM-32B, in the manifest order, keeping Bolt.diy, the memory pipeline and the evaluators unchanged. The 32B model is served in BF16, as in the original paper, where it reaches 38.2% with Bolt.diy [19]; the two arms run concurrently on two lanes, so both experience the same infrastructure conditions. Deployment differs from the 7B runs: the 32B generator shares the second GPU with the memory model, which is restarted with a smaller memory share (0.42 of the GPU instead of 0.80, with the 32B model given 0.54); the evaluator and the 7B lanes stay on the first GPU. This comparison shows whether the full memory helps a stronger generator; it does not by itself show whether a larger model makes better use of text than of code, because the 32B runs have no text-memory arm. That question is left to future work.

[TODO: Status: 32B runs scheduled for the weekend of 10-17 on the second GPU. Table: 32B No Memory vs Final (v9.1) in the format of Table 5.1, with the paper's 32B row; paired difference with CI; and the share of zero-score tasks.]

## 5.7 Robustness to Task Order

WebGen-Bench is a static benchmark: tasks are solved independently and its main results are single runs without error bars, which the authors attribute to computational cost [19]. With memory, however, the order of tasks determines which experience is available, so a single order is not sufficient evidence. In particular, in the manifest order tasks of the same domain are adjacent, so a later task can always draw on a just-solved, similar website; a gain could therefore reflect a curriculum effect rather than general learning.

We therefore compare two kinds of order, each run twice, every run starting from an empty memory:

**Domains adjacent.** The order of the 20 domain blocks and the order of tasks within each block are shuffled, with seed 11 and seed 47. The manifest order, used for the main results, also belongs to this kind.

**Domains mixed.** A uniformly random order of all 101 tasks (seed 29), and an interleaved order that takes one task from each domain in turn, so that consecutive tasks always come from different domains.

In every order, Long Memory and No Memory are run as pairs on each task, alternating which arm runs first (ABBA), so that drift in infrastructure or evaluator load affects both arms equally. No Memory should not depend on order; its runs instead measure run-to-run noise (three completed, two running). The checklist-only ablation does not use memory and is run once. Results are task-level paired differences with bootstrap CIs.

**Development and validation.** Seed 11 was used to develop v8–v9.1 (Section 5.7.1) and is therefore development data for the final version. The manifest, seed-29, seed-47 and interleaved runs of v9.1 started after v9.1 was frozen. All orders use the same 101 tasks, on which the earlier versions were developed, so this is a within-benchmark study with cross-order validation, not a held-out evaluation (Section 6.4).

The first two shuffled orders give almost the same result (Table 5.5): Long Memory leads by about 4.5 points, with zero-score tasks reduced from 29 to 21, but neither confidence interval excludes zero, and the gain is far smaller than in the manifest order. No Memory is stable across its three runs (27.9–28.6%), so the change is on the Long Memory side. Mixing domains does not remove the gain: the random order (seed 29) matches the shuffled-blocks order (seed 11), and from task 31 onwards Long Memory

Table 5.5: Task-order experiments: accuracy (%) of No Memory (NM), v7 and the final version v9.1, paired difference to No Memory with 95% bootstrap CI, and zero-score tasks. Single run of each arm per order. [TODO: Status: manifest (v9.1) being scored (10-16); seed 47 running; interleaved queued. Fill in when complete.]

| Kind | Order | NM | v7 | v9.1 | v9.1 − NM | Zero NM / v7 / v9.1 |
|---|---|---|---|---|---|---|
| Adjacent | Manifest[a] | 28.5 | 40.0 | – | – | 30 / 13 / – |
| Adjacent | Shuffled blocks (seed 11) | 27.9 | 33.1 | 35.4 | +7.4 [1.6, 13.1] | 29 / 21 / 16 |
| Adjacent | Shuffled blocks (seed 47) | – | – | – | – | – |
| Mixed | Random (seed 29) | 28.6 | 32.6 | 33.7 | +5.5 [−0.5, 11.6] | 29 / 21 / 18 |
| Mixed | Interleaved | – | – | – | – | – |

v7 − NM: manifest +11.5 [6.0, 16.9]; seed 11 +4.4 [−1.9, 10.6]; seed 29 +4.5 [−2.2, 11.3].

[a] Not run as alternating pairs; the two arms were run at different times.

leads by 6.6 points in the random order (33.9% vs 27.3%). A curriculum effect from same-domain neighbours is therefore not the main source of the benefit.

Both runs lose much of their gain to the same failure (Figure 5.3). Long Memory returned the unmodified Vite scaffold on 19 tasks in seed 11 and 16 in seed 29, against none for No Memory in either order and two in the manifest order. Every scaffold occurred after the missing-package code memory, whose example was a complete `App.vue` of another site, began to be injected (task 36 in seed 11, task 19 in seed 29). This is the failure the v8 change targets (Section 5.7.1).

With the failure fixed (v9.1, Section 5.7.1), no scaffold occurred in either order (19 → 0 and 16 → 0) and the gain over No Memory rose to 7.4 points in seed 11 (95% CI [1.6, 13.1]) and 5.5 points in seed 29 ([−0.5, 11.6]), with zero-score tasks reduced from 29 to 16 and 18 (Table 5.5 and fig. 5.3). Positive gains were observed with domains adjacent and with domains mixed, although the interval for the mixed order includes zero; on this evidence the gain does not rely on same-domain neighbours. Compared directly with v7, v9.1 is ahead by 3.0 and 1.0 points, which is within the run-to-run variation of a single run; the evidence for the fix is the removal of the scaffold failures, not a higher total. Missing Vue imports remain (23 and 27 tasks with at least one affected component, against 17 and 19 for v7): once the generator writes real pages instead of the scaffold, it has more opportunities for this slip.

[TODO: Status: manifest being scored, seed 47 running, interleaved queued. Report whether the gain holds in all orders and whether the interleaved order is clearly worse; add one summary figure of the gain and CI per order, with a shared y-axis across the order plots.]




Figure 5.3: Rolling 10-task mean of the per-task accuracy gap to the No Memory run of the same order, by execution order. Orange: code memory with a whole-file example (v7); blue: verified fragments only (v9.1). Positive values mean memory is better. The v7 dip after task 45 in seed 11 coincides with the scaffold outputs.

[TODO: Planned (existing data, final version): (1) a per-domain table of the gain over the 20 domains; (2) memory use over time: entries injected per task, injected tokens, size of the memory, first activation of each code-pool entry; (3) one complete case of transfer (failure, reflection, stored entry, why it was retrieved, injected text, change in code, outcome) and one of negative transfer (Figure 4.1).]

### 5.7.1 Fixing the Exposed Failure

The fix was developed on seed 11 while the frozen v7 was used for all other experiments. It addresses three properties of the code pool in turn.

*Form of the example.* The scaffolds came from the one entry type that still showed a whole file. v8 therefore stores a fragment for every entry type. On the seven seed-11 tasks where v7 had returned the scaffold, accuracy rose from 3.6% to 45.2%, and no scaffold occurred in the 61 tasks of the partial run. Total accuracy on the other tasks is not informative here: their prompts already differed before the code pool, because

memory chains diverge from the first task, and two runs of the same version can differ by about five points. Versions are therefore compared by mechanism counts.

*Stability and coverage.* In v8 the missing-import example was rebuilt whenever the counts of missed functions changed, and it switched between two versions. v9 replaces an entry only when the new example covers strictly more of the needed functions. It also creates entries for whole-site failures after one occurrence, because waiting for a second occurrence meant that rare problems were learned after they had already occurred, and it allows three slots instead of two.

*Applicability.* The compile-error entry of v9 carried an applicability condition that no requirement matched, so it was never injected. v9.1 applies compile-error entries to every Vue site and is the final version.

The evidence for these changes comes from small targeted checks and partial runs on seed 11; it is debugging evidence. The full-benchmark evidence for the final version is the set of runs started after it was frozen (Table 5.5).

# Chapter 6

# Discussion

## 6.1 The Role of Memory Representation

The central observation of this work is that this small generator used memory very selectively. In diagnostics, single textual sentences recovered at most one blank-page task in three where a short code example recovered seven of nine (Section 4.2); in full runs, generic lessons accumulated in the text-dominated memory, and short code fragments were followed. v4 and v7 differ in more than the form of memory, so this is an observation across system versions, not a controlled comparison of text and code; the decline of v4 is consistent with interference from accumulated text, but the experiments do not isolate that mechanism. This agrees with Metis's finding that code memory complements text memory [5], and adds that for a 7B generator the format of the memory can decide whether it helps at all. The form of the code also matters: a fragment that looked like a complete file was copied as a finished answer, producing an untouched scaffold.

This contrasts with strong models, for which text memory transfers more reliably than code [5, 15]. The mechanisms differ (Metis executes its code; we show code to the generator), but together the results suggest that the best form of memory depends on the capability of the model that consumes it.

A second observation is that part of the gain is not "smarter" code but protection against collapse. In the earliest phase, short prompts made the model return the default scaffold in five of eight tasks, while longer, specific prompts did not. Both effects are real benefits of memory, but they should be described separately.

## 6.2 What Is Learned

The generator does not change; what changes is which reusable code is shown to it, and when. The system learns from its own outcomes which problems recur and which of its own earlier code avoids them, and it stops injecting lessons that do not prevent a recurrence. This is closer to automatically building a small, evidence-backed set of safeguards than to acquiring new programming ability, and it is a property of the whole system, including a memory model larger than the generator. Within 101 tasks the memory stays small (a few code-pool entries and tens of active lessons); how it behaves over thousands of tasks, with many more problem types competing for three slots and older entries conflicting with newer ones, is not tested here.

## 6.3 Cost

The generator is small, but the system is not. Each Long Memory task takes about 12 minutes against about 6 for No Memory, mainly for checklist construction, applicability judgements and reflection by the 72B memory model. The functional tester is shared by both arms and belongs to evaluation, not to deployment. If the fixed-prefix ablation shows that the learned fragments alone recover most of the gain, a deployed system could run the 72B model offline and ship only the fragments. [TODO: Planned (existing logs): cost table per task (time, memory-model tokens, GPU memory), separating evaluation cost shared by both arms from the extra cost of memory.]

## 6.4 Limitations and Threats to Validity

- **Single runs.** WebGen-Bench itself reports single runs without error bars. We repeat No Memory under several orders to measure run-to-run noise (three runs completed, two running) and repeat Long Memory under additional orders with paired comparisons, but within each order Long Memory is still a single run. Two runs of the same Long Memory version can differ by about five points, more than No Memory varies, so the task-level intervals understate the full uncertainty.
- **One generator and one scaffold.** All results are for WebGen-LM-7B with Bolt.diy and a Vue template. The finding that code is followed where text is not may be specific to this model rather than to small models in general; larger generators may follow text well and benefit less from code fragments. The 32B run

(Section 5.6) tests whether the memory also helps one larger model of the same family, but not whether that model prefers text.

- **Residual generator errors.** About 4.6% of generated components omit a Vue import with or without memory; examples did not reduce it, and the protocol forbids repairing it after generation.
- **Code pool as a learned prefix.** Because its triggers are general, the code pool behaves partly like a fixed prompt prefix; the fixed-prefix ablation (Section 5.5) measures how much of the gain this explains.
- **Rare problems are learned late.** A code memory needs a problem to recur before it exists, so learning pays off mainly for frequent problems. In v7 the `vue-router` and `v-model` problems each occurred in only two seed-11 tasks, and with two slots the `v-model` memory was never injected although later tasks failed with that error. v9.1 creates memories for whole-site failures after one occurrence and allows three slots, but a problem seen once can still cost one task, and with other generators more problem types would compete for the slots.
- **Appearance grader.** We grade appearance with Qwen2.5-VL-32B instead of GPT-4o, so appearance scores are compared only between our arms. On tasks where both arms produce a working site, mean appearance is similar (3.25 vs 3.22); this subset is selected by the outcome, so it does not establish that memory has no effect on appearance.
- **Iterative development on the same benchmark.** This is the most serious threat. Every version was designed after looking at failures on the same 101 tasks on which it is evaluated, so the design is in effect tuned on the test set. Four facts limit, but do not remove, the risk. Reflection rules teach a method of investigation and never a task-specific answer; memory content is learned from earlier tasks only and every run starts empty; the split into the first 30 and the last 71 tasks was fixed before v7 was run; and the final version was run after it was frozen on the manifest order and on shuffled orders that were not used to develop it (only seed 11 was). Because the 30/71 split was also seen during development, it cannot serve as a held-out test after the fact; the results should be read as a within-benchmark study. A clean estimate would freeze the method on a development subset and report it once on held-out tasks; we leave this to future work (Chapter 7).

- **Evaluator noise.** GUI testers can exhaust their interaction budget or misjudge pages [37]. We keep raw evaluator transcripts and distinguish interaction-budget exhaustion from logic failures. [TODO: Planned: manual check of 20 sampled test cases (stratified by Yes/Partial/No and by arm); report agreement between the tester and a human reader.]

- **Memory model stronger than generator.** The 72B memory model is larger than the 7B generator, so part of the gain could be the larger model solving the task for the small one. Three observations argue against this reading. The checklist only restates the requirement, and on its own its gain has an interval that includes zero (Table 5.4). Every code fragment is copied verbatim from a website that the 7B generator itself built earlier, and code checks reject lines that are not in that file. And the memory model never sees the current task's evaluation. The results therefore show what a small generator gains when a stronger model organises its own past experience, not self-improvement in isolation. [TODO: Status: fixed-prefix run queued after seed 47. State how much of the gain remains without accumulated memory and without the 72B model at generation time.]

## 6.5 Ethical Considerations and Impact

**Data and privacy.** The benchmark requirements are synthetic and contain no personal data. All models run on university hardware; no task data were sent to external services.

**Environmental cost.** Each full run of 101 tasks takes about a day on two H200 GPUs, and the study required many runs. We reduced cost by testing changes on three to five tasks, by checking generated code statically before evaluating, and by avoiding repeated validations.

**Self-improving code agents.** A memory that turns observed outcomes into future behaviour can also entrench mistakes or unsafe patterns. Our design keeps full provenance of every memory, requires code examples to compile, and never edits the source record, so that any injected content can be traced back to the task that produced it. Deployed systems would additionally need review of stored code for security.

**Economy and society.** Tools that let small, locally hosted models build working websites lower the cost of software creation and keep data on local infrastructure, which

may benefit small organisations. They may also reduce demand for routine front-end work. Generated websites still require human review for correctness and accessibility.

**Use of AI tools.** [TODO: To confirm with the supervisor: course policy on the AI-use declaration.] AI assistants were used for programming, organising experiment logs and editing English. The research questions, experimental design, analysis and conclusions are the author's own, and every number in this dissertation was computed from the archived runs.

**Research integrity.** We report negative results (short CoT, the scaffold regression in v6, v4's decline) and invalidated our own earlier results when a delivery bug was found.

## 6.6 Project Management

The project was organised in versions. The cluster ran one frozen version at a time; new ideas were tested locally on small samples and with static code checks, and a version was replaced only when the new one was shown to be better. Decisions and their evidence were recorded continuously in a decision log, which made it possible to reconstruct why each component exists.

# Chapter 7

# Conclusion

## 7.1 Answers to the Research Questions

***RQ1*** A qualified yes, for WebGen-LM-7B with Bolt.diy on WebGen-Bench. In a single run over 101 tasks v7 improves accuracy from 28.5% to 40.0% (paired gain 11.5 points, 95% CI 6.0–16.9) and reduces zero-score tasks from 30 to 13 under the official protocol, without changing the generator or the evaluation. The gain is in functional accuracy; much of it comes from sites that now render at all. [TODO: Status: replace with the final version (v9.1) on the manifest order (being scored, 10-16), and summarise all orders once seed 47 and interleaved finish.]

***RQ2*** In this system, the small generator followed short code fragments copied from its own earlier websites. A requirement checklist alone reduced whole-site failures but added only 3.4 points of accuracy (interval includes zero), while memory added 8.1 points on top of it (interval excludes zero). Textual lessons were rarely followed in diagnostics, and the text-dominated version fell behind No Memory late in the sequence; because the versions differ in more than the form of memory, this is not a controlled comparison of text and code. Fragments must be partial: a complete file was copied as a finished answer. [TODO: Status: add the fixed-prefix result (queued after seed 47) and the 32B result.]

***RQ3*** The v7 advantage grows over the sequence (40.4% vs 29.5% on tasks 31–101, 38.3% vs 26.0% on the last 35), while v4 falls below No Memory after task 30, consistent with, but not proof of, learning from accumulated experience; the fixed-prefix ablation tests the alternative of a fixed prompt. In two shuffled orders v7's gain fell to about 4.5

points because one code memory, a whole-file example, made the generator return the scaffold; with examples reduced to verified fragments (v9.1) the scaffolds disappeared and the gain was 7.4 points with domains adjacent (interval excludes zero) and 5.5 points with domains mixed (interval includes zero). Positive gains were observed in both kinds of order, but robustness to order is not yet established. [TODO: Manifest, seed 47 and interleaved order with v9.1.]

## 7.2 Future Work

Two further experiments would address the main remaining questions:

- **Held-out evaluation.** Because the method was developed while observing results on all 101 tasks, a clean test needs tasks that were never seen: a held-out domain, or a second website-generation benchmark, with the method frozen beforehand.
- **Controlled form-of-memory comparison.** Keep retrieval, trigger and token budget fixed and vary only the form: textual rule, pseudocode, short fragment, complete file, plus random-code and length-matched text controls.

Further directions are integrating the memory with other coding agents (e.g. OpenHands, Aider) on a stratified subset of tasks, to show that the approach does not depend on Bolt.diy; re-grading with a second strong VLM evaluator; repeating each order more than once to separate order effects from run-to-run variation; letting the memory model revise reflections and fragments until code-checkable criteria hold [20, 7], and transferring the approach to repository-level coding benchmarks such as SWE-bench.

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